

How to Approach These Questions

The assessment and interview include guesstimate and case-study prompts. Interviewers value structured thinking and explicit assumptions over an exact final number.

Framework: clarify, break into MECE parts, state assumptions, compute step by step, and sanity-check.

Guesstimate Questions (structured approach)

Prompt 1. Estimate the number of smartphones sold in India in a year.

Model approach: Population ~1.4B; ~50% own a smartphone (~700M users); ~3-year replacement cycle => ~230M replacements/year; add ~30-50M first-time buyers => ~250-280 million units/year. State each assumption and reason top-down.

Authenticity: Community Reported

Prompt 2. Estimate cups of tea sold in a mid-sized Indian city per day.

Model approach: City ~2M people; ~40% drink tea outside home (~800k) at ~2 cups/day = ~1.6M; add office/vendor demand => ~1.5-2 million cups/day. Segment by consumer type.

Authenticity: Community Reported

Prompt 3. Estimate the annual revenue of one multiplex cinema screen.

Model approach: ~5 shows/day x ~150 seats x ~40% occupancy = ~300 tickets/day; ~Rs.200/ticket = ~Rs.60k/day; add ~50% from F&B => ~Rs.3.15 crore/screen/year.

Authenticity: Research-Backed Company Style

Prompt 4. Estimate daily data (GB) generated by a ride-hailing app in one city.

Model approach: ~100k rides/day; ~200 KB structured data/ride (GPS, IDs, payment, ratings) => ~20 GB/day core; add telemetry/logs for a higher figure. State the data types counted.

Authenticity: Research-Backed Company Style

Case Study / Analytical Prompts

Prompt 1. A retail client's online sales dropped 15% last month. How would you investigate as a data analyst?

Model approach: (1) Verify the data (tracking, seasonality, one-offs). (2) Segment the drop by channel, region, device, category, customer type. (3) Compare funnel metrics (traffic, conversion, AOV). (4) Check external factors (competitor promos, pricing, stockouts, outages). (5) Quantify each hypothesis and recommend actions with expected uplift.

Authenticity: Community Reported

Prompt 2. A subscription business wants to reduce churn. What analysis and metrics would you propose?

Model approach: Define churn and window; build cohort/retention analysis and CLV; identify drivers via segmentation and a classification model (logistic/gradient boosting) using tenure, usage, support tickets, payment issues; rank feature importance; target high-risk segments; measure impact via an A/B test.

Authenticity: Community Reported

Tredence

Guesstimates & Case Studies

Applied Role(s):

Analyst - Data Scientist

Online Assessment Preparation Material

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Every question carries an authenticity label. See the legend on the next page.

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Authenticity Legend & How to Read This Document

Every question and problem in this document is labelled with an **authenticity classification** so you know how closely it maps to what candidates have actually reported. Labels are assigned conservatively based on research; authored practice questions are never presented as real exam questions.

Label	Meaning
Verified Previous-Year	The specific problem/pattern is documented from actual reported assessment experiences.
Frequently Repeated	The topic or pattern type is documented as recurring across multiple hiring cycles.
Community Reported	A widely shared, role-relevant pattern reported across placement communities and forums.
Research-Backed Company Style	Authored to match the researched difficulty, style and pattern - practice, not a real past question.

The correct option in every MCQ is highlighted in green and also restated below the question. Study the pattern, not just the answer.